

Team Name: CImEx Compliance

Team Size: 4

Problem Statement:

From 2026 the EU charges for the carbon in imported steel goods. Indian exporters without verified data get priced on penalty defaults, and most cannot produce that data from the paperwork they have.

Problem Statement

The challenge or problem identified by the team.

- **Why now.** The EU carbon border tax (CBAM) became a real cost in 2026. The payable share rises from **2.5% in 2026 to 100% in 2034**. The first annual declaration is due **30 Sep 2027**.
- **The penalty.** Suppliers without verified data are priced on EU default values plus a **10% to 30% mark-up**. Indian goods become more expensive than rivals with data.
- **The scale.** India sold **€5.30 bn** of CBAM-covered goods to the EU in 2025.[1]
- **Why it is hard.** The data sits in scanned mill certificates, invoices and fuel bills. Experts are scarce and costly. Most of a fastener's number is its steel, and only the mill has that data.
- **Who feels it.** Mid-size Indian metal exporters: fasteners, wire and pipes. We start in Chennai and Hyderabad.

1. Eurostat Comext, via CBAMGuide.

Proposed Solution

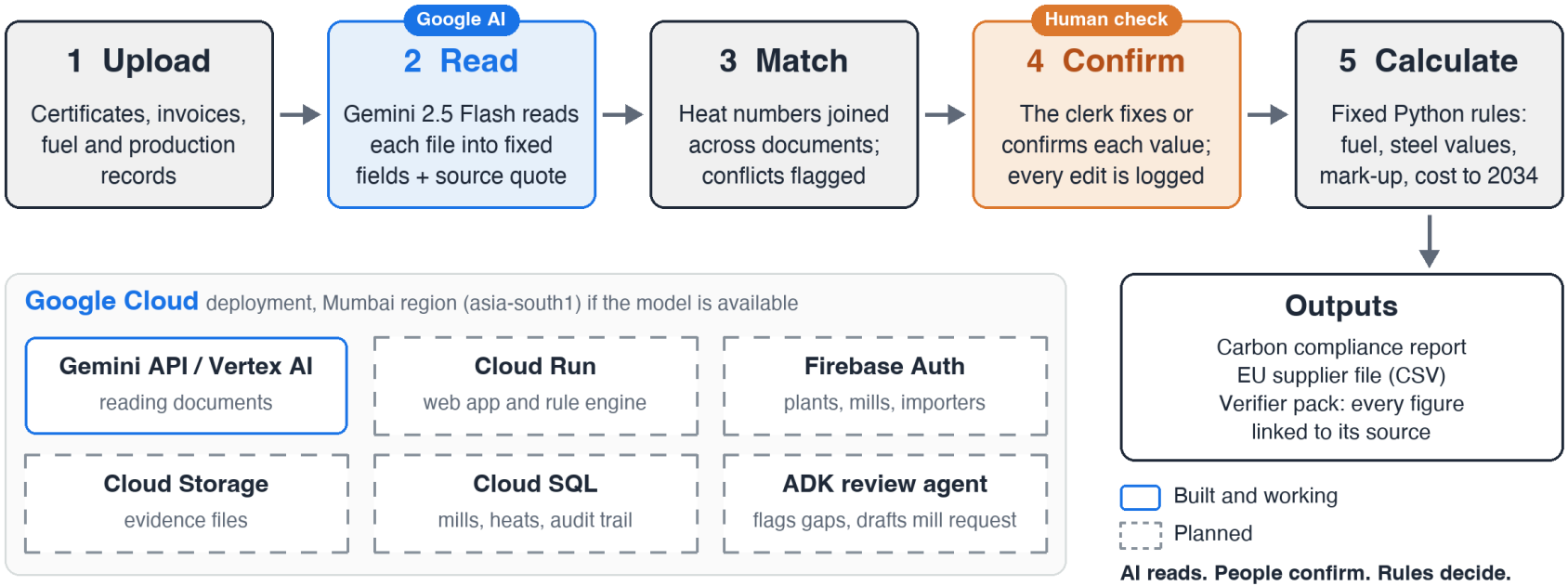
The idea and how it addresses the identified problem.

ClmEx Compliance turns a plant's existing paperwork into carbon data an EU verifier can check. The clerk who already handles the paperwork runs it.

- **How it works.** Upload certificates, invoices, fuel and production records. Gemini reads each one and shows the source line. The clerk confirms or fixes every value. Fixed rules calculate. Out comes a report, an EU supplier file and a verifier pack.
- **Evidence, not estimates.** Every figure traces to a line in a document, and every correction is logged.
- **AI reads, rules decide.** Gemini never sets the number, so the same input always gives the same result. Verifiers can trust that.
- **Network effect.** Verify a mill once, and every plant that buys its steel moves off the penalty default. In our demo, one mill sharing data takes a plant from **4.40 to 2.25 tCO2 per tonne**.
- **Working today.** A live demo runs the full flow from upload to report.

Solution Architecture

Architecture of the proposed solution



Google AI Tools

Google AI Studio, Google Antigravity and Google Agent Skills (used or planned to be used).

- **Gemini API** **Used**
  Reads scanned certificates, invoices and fuel bills into a fixed set of fields, with the source quote for each value.
- **Google AI Studio** **Planned**
  Test prompts and structured-output schemas on real certificate layouts from pilot plants, and measure accuracy per field.
- **Google Antigravity** **Planned**
  Build the hosted app on Cloud Run and the review agent during the Buildathon.
- **Google Agent Skills / ADK** **Planned**
  A pre-verification agent with skills for CBAM rules, heat matching and gap checks. It flags what a verifier would reject and drafts the data request to the mill.

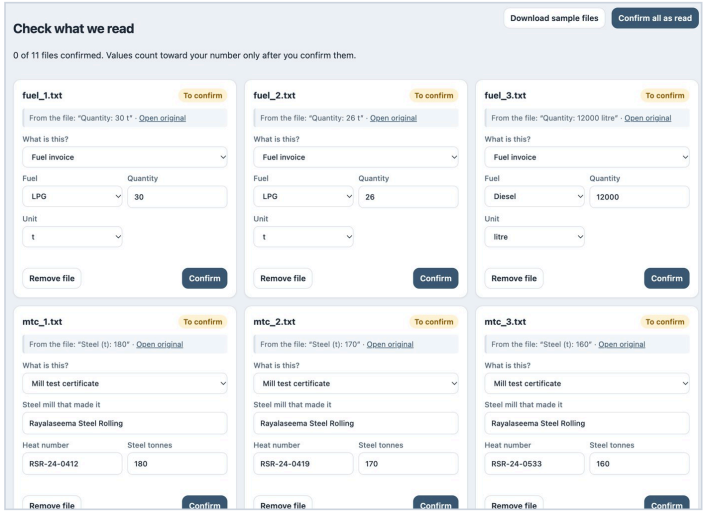
Google AI Models

Google AI model(s) used or planned for the proposed solution.

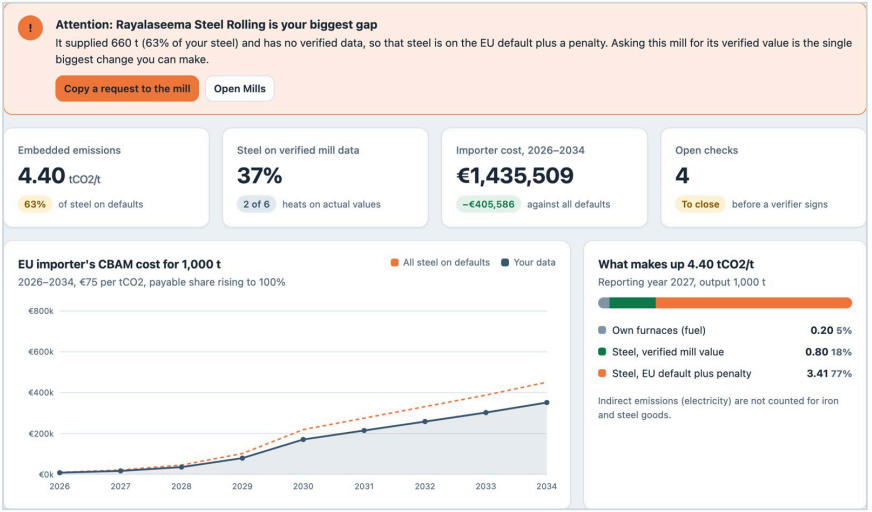
- **Gemini 2.5 Flash** Used
  Multimodal and low cost per page. It is the model set in our code. It must read scans, stamps, phone photos and Indian number formats such as 1,00,000 kg.
- **Structured output.** Gemini returns fixed fields: mill, heat number, tonnes, fuel type and quantity, plus the source quote. Our rules reject impossible values, such as negative tonnes.
- **Measured accuracy.** Every clerk correction is logged, so real documents give a field-by-field accuracy score for Gemini.
- **Vertex AI** Planned
  Move to Google Cloud, in the Mumbai region if the model is available there.
- **Gemini 2.5 Pro** Planned
  Test it on hard scans if Flash accuracy is too low.

Working prototype, live today

https://hemanthdadi1.github.io/AI-Export-Compliance/



Review: each value with its source line, to confirm or fix



Dashboard: 4.40 tCO2/t, the biggest gap, and the buyer's cost to 2034

Built and tested: document reading, review and confirm, heat matching, rule engine, report, verifier pack, mill network view. Sample data is made up.

Next 30 days: real plant documents through Gemini, hosting on Cloud Run with Vertex AI, and three pilot plants.

Google DeepMind | Google Cloud

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Thank you

